

Role-based network embedding via structural features reconstruction with degree-regularized constraint

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ABSTRACT

Role-based network embedding aims to map network into low-dimensional node representations while preserving structural similarities. Adjacency matrix contain both the local and global information of a network, but it cannot directly represent the role of nodes. So it is essential to extract higher-order structural features from adjacency matrix for role-based network embedding (structural equivalence). While being sensitive to noise in real networks in general, the features extracted by some identical methods cannot truly represent the role of nodes. Therefore, we propose a deep learning framework RESD. In detail, we first propose extracting higher-order structural features for each node in the network. Then, we utilize the Variational Auto-Encoder (VAE) to model the nonlinear relationship of the features, reduce noise and improve the robustness of embedding. Furthermore, in the embedding space, we apply a degree-regularized constraint to guide the representation learning for preserving key structural information of nodes (e.g., degrees), which may be lost due to the principle of VAE framework. Finally, we construct a unified objective function to learn the node embedding for role discovery by preserving the structural features and node degree. We compare our model with several state-of-the-art methods on real-world networks. The results of extensive experiments demonstrate the effectiveness of our model and prove that our model scales well with dimension and network size.

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1. Introduction

Graph, or network, is a generic structure to represent irregular data, where nodes denote real-world entities and edges between them denote their relationship. Network structure is so universal that a variety of complex systems can be modeled by it, such as social networks [1] and protein networks [2]. In recent years, network embedding (NE) has aroused considerable interests of researchers [3–5]. It aims to learn low-dimensional representations of nodes or the whole graph while preserving its structure [6,7], which can be applied into a lot of downstream tasks, including node or graph classification [8], community detection [9], social recommendation [10,11], and link prediction [12,13].

Nearly all these NE methods are designed to preserve node proximity, which usually leads a clustering guided by communities in the network. As shown in Fig. 1, the parts surrounded by the dashed lines are two different communities separated based on the tightness of connection. The popular NE methods, such as DeepWalk [14] and node2vec [8] which are based on random walks, can effectively extract node proximity and within-cluster

density. Based on node embeddings, it is easy to cluster nodes as a community, as nodes surrounded by red or blue dashed lines. Similarly, there are also many other methods consisting of GCN [15] and GraphSAGE [16] which diffuse features through the network to achieve this. These methods are usually known as community-based network embedding.

However, role, another important point in complex networks, is fundamentally different to the notion of community. The definition of role is first presented as classes of equivalent nodes [17], which are used to present the structural similarity among nodes. In other words, roles are node partitions satisfying the criterion: nodes are more structurally similar to nodes inside the same partition than those outside. Intuitively, roles represent the connection patterns of nodes in their neighborhoods. As shown in Fig. 1, the two red nodes share the same role, for they can be regraded as bridge of two sub-graphs, and the yellow nodes have the same role because they are both the center, even if they are far from each other or unconnected. Hence, role-based network embedding is designed to project the network into embedding space in which nodes of the same role are closer together. To sum up, although these two problems are devoted to the clustering of nodes in networks, they have essential differences and their respective priorities [18].

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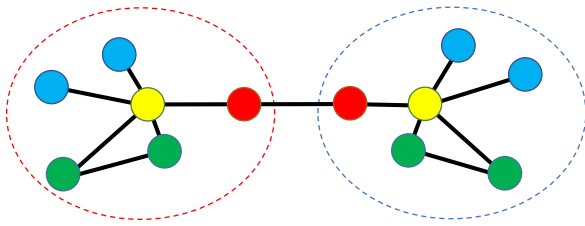


Fig. 1. An illustration of roles and communities in a network. In this figure, nodes in the same color denote that they share the same roles, while the circles of dashed line denote the communities.

Methods for community-based network embedding are usually failed to solve the role-based task for they cannot capture and model the structural characteristics of nodes. Recently, some methods have been proposed to capture structural similarity and obtain role-based network embeddings. One of the most representative way is based on random walks. The first challenge for role discovery is that nodes in the same role may not be directly connected or far away from each other. To address this problem, some methods improve the ordinary random walks. For example, role2vec [12] adopts a feature-based random walks instead of id-based random walks to generate sequences as input of SkipGram [19]. However, it is still powerless to deal with extracting structural identities for distant nodes, and cannot be applied to large-scale networks.

Some methods attempt to leverage the adjacency matrix rather than random walks to alleviate the problem of non-connectivity between nodes in the same role. However, the adjacency matrix of network could contains first-order and high-order relationships, it cannot directly present the structural similarities. So the adjacency matrix is inapplicable to be simply used to generate role-based embeddings. Then some methods turn to extract structural features based on adjacency matrix for role discovery. For example, ReFeX [20] is an effective method that recursively aggregates the neighborhood features and egonet-based features. It has a good performance and obtains a relatively compact representation. But with the increase of the number of iterations, each node can meet fairly high-order neighbors, which may cause over-fitting and result in that the features cannot truly present structural information. RoIX [21] and GLRD [22] leverage ReFeX features and decompose the feature matrix to obtain compact node representations.

Nevertheless, the methods based on decomposing matrix may cause high computational complexity. What is more, the features they used depend on the neighborhood information, which make them extremely sensitive to noises in networks, and the embeddings generated by them lose some statistic features of nodes, such as node degree. Inspired by deep learning, DRNE [23] leverages a layer-normalized LSTM regularized by node degree to recursively aggregate neighbor's representation to capture structural equivalence. This mechanism can effectively capture structural similarity between distant nodes, but it still ignores the noises.

Above all, to learn role-based node embeddings, we must solve challenges of non-connectivity and noise while preserving statistic features of nodes. Thus, we propose a deep learning framework called **Role-based network Embedding via Structural features reconstruction with Degree-regularized constraint (RES_D)**. The framework consists of three parts: feature extraction, embedding generation and topological characteristics preserving. In our framework, we first extract higher-order structural features for each node. In this paper, we choose ReFeX as a sample feature extraction method for its efficiency and effectiveness. Then, we leverage Variational Auto-Encoder (VAE)

[24] on feature matrix to reduce noise and obtain more compact representations, it can also capture nonlinear relationship between input and node embedding, which has well robustness and avoids over-fitting. However, the encoding process of VAE may lead to the loss of some key structural characteristics of the nodes. Therefore, the regularization is added on VAE to preserve node degrees, which plays an auxiliary role to guide the direction of optimization.

Our main contributions can be summarized as follows:

- We propose a novel framework for generating role-based network embedding via structural features reconstruction. This process can obtain a more compact node representation and improve the performance of features.
- We leverage Variational Auto-Encoder on features to capture nonlinear relationship between input and node embedding while preserving structural similarity. It can also reduce the noise in original networks and increase the robustness of representation.
- The process of encoding loses structural features (e.g. degree) and we confirm that the node degree is significant for the distribution of roles. So we add degree-regularized constraints on VAE to preserve structural identity and enhance performance.
- We conduct several extensive experiments on real-world networks via our model *RES_D*, and compare the results with other state-of-the-art methods. The results demonstrate the superiority of our model, and prove that our model scales well with network size and dimension.

The rest of the paper is organized as follows: Section 2 proposes an overview of related works for role-based network embedding.

Section 3 introduces the proposed model *RES_D* and the process for generating representations in detail. In Section 4, a variety of real-world networks and experimental results are provided, and we conclude the paper in Section 5.

2. Related work

The role in networks indicates the structural identities of a node, which is reflected as its local connective pattern. Role discovery has been widely studied over the past decades.

Traditional studies generally define roles through graph representation. The most popular class of methods among them is the blockmodels, which construct a framework that nodes represent roles and edges represent interactions between roles. CONCOR [25] is the first to compute the correlation of adjacency matrix in a recursive way. Stochastic blockmodels (SBM) [26] is a classical method which introduces stochastic equivalence into role identity, thus nodes are in the same role if the probability of linking to all other nodes are the same. Then some methods improve SBM to discover roles. For example, mixed-membership stochastic blockmodel (MMSB) [27] leverage the idea of blockmodels and assumes that one node can belong to multiple roles.

Other traditional methods leverage the mechanism of feature engineering, which shift the focus from graphical representations to node representations. They try different measurements to evaluate some particular structural identities of nodes, such as degree centrality, eigenvector centrality [28], PageRank centrality [29], and structural hole value [30], which may show good effectiveness on some special tasks. ReFeX [20] is an effective method to capture structural features. It recursively aggregates the neighborhood features and egonet-based features to obtain global structural information. But they are still hard to be applied to discover node roles and complex tasks as too limited structural information is preserved.

Recently, to capture comprehensive role information and leverage it to complex machine learning tasks, researchers have tried to take advantage of network embeddings. Most of these role-based network embedding methods can be divided into three categories based on the embedding mechanisms: methods based on languages models, matrix factorization and deep learning.

The first type of methods are based on the language model Skip-Gram [19] applied on random walks. Classic random-walk-based network embedding methods such as DeepWalk [14] and node2vec [8] are all tied to community because of their id-based random walks, but they are not applicable for role discovery. Some methods have attempted to improve random walks so that it is applicable to role detection. The most effective model of this kind for role discovery is struc2vec [31]. It computes multiple node-to-node feature-based similarities between all pairs of nodes using dynamic time warping (DTW) distance, and then constructs a hierarchy of complete graphs whose edges weights are transformed from the feature-based similarities and used for computing probabilities of random walks. More recently, role2vec [12] applies random walks on motif-based features, and node2bits [32] has applied the same idea for tasks of user stitching, and also leverages a temporary walk CTDNE [33]. RiWalk [34] designs new random walk methods to approximate graph kernels on subgraphs.

The mechanism of matrix factorization have been widely applied for role-based network embedding. Most existing methods construct a feature matrix or a similarity matrix, and then compute compact node representations via matrix factorization. RolX [21] is the first to leverage this mechanism to generate role-based node embeddings. It leverages the structural features extracted by ReFeX [20] and use matrix factorization to get low-dimension node representation. It is demonstrated that this model has great potential for role discovery. Later, GLRD [22] takes advantages of these features and xNetMF in REGAL [35] also constructs a feature matrix and decompose it, while HONE [36] generates high-order network embeddings by decompose a set of motif-based matrices. They can effectively extract the structural similarity between nodes. Some methods leverage classical approaches to compute pair-wise similarity. REACT [37] applies RoleSim [38] matrix to get similarity between each pair of nodes, and then obtain node representations. In addition, SEGK [39] decomposes the similarity matrix computed by graph kernels. Besides above methods, GraphWave [40] generate embeddings by applying graph diffusion kernels and treating them as probability distributions over the graph. Another work [41] designs four multilayer network embedding algorithms based on Nonnegative Matrix Factorization (NMF) while preserving structural properties. But methods of this kind based on matrix factorization are not applicable for large-scale networks except RolX because of their high computational and spatial complexity.

Deep learning frameworks based on neural network have been widely applied in many areas of researches, such as image classification [42] and natural language processing. Recently, some deep learning methods are proposed for network embedding to capture the highly nonlinear [43] relationship between topological structure of network and node representations. DRNE [23] applies a layer-normalized LSTM [44] to learn regular equivalence. It samples the neighbors of each node and sorts them by node degree, then uses the representation of neighbors as inputs to LSTM regularized by a multi-layer perceptron to reconstruct node degree. Inspired by it, other deep learning methods appear. Demo-Net [45] constructs a degree-specific GCN, which uses different parameters with respect to node degree, but it is for semi-supervised condition. DMER [46] leverages a GCN encoder on adjacency and an Auto-Encoder on ReFeX [20] features, and GAS [47] also leverages GCN on simple structural features to generate node embeddings. However, these deep learning methods are still sensitive to noises.

Table 1

Symbols and Definitions.

Symbol	Definition
n	The number of nodes
e	The number of edges
m	The number of feature dimension
k	The number of embedding dimension
L	The number of layers
$d(v_j)$	The degree of node v_j
$\mathbf{x}_j$	The input feature of node v_j
$\mathbf{h}_j^l$	The l th encoder layer's representation of node v_j
$\hat{\mathbf{h}}_j^l$	The l th decoder layer's representation of node v_j
μ_j	The mean value of the distribution of node v_j
σ_j	The standard deviation of the distribution of node v_j
ϵ	The random variable from Standard Gaussian Distribution
$\mathbf{z}_j$	The representation vector of node v_j
$\mathbf{Z} \in \mathbb{R}^{n \times k}$	The representation matrix of network
$\mathbf{W}^l, \mathbf{b}^l$	The weight matrix and biases of the l th encoder layer
$\mathbf{W}^\mu, \mathbf{b}^\mu$	The weight matrix and biases calculating μ
$\mathbf{W}^\sigma, \mathbf{b}^\sigma$	The weight matrix and biases calculating σ
$\hat{\mathbf{W}}^l, \hat{\mathbf{b}}^l$	The weight matrix and biases of the l th decoder layer
β	The weight of degree-based regularizer.

3. Method

In this section, we will first declare the notions used in this paper, and then introduce our framework *RESD* in detail.

3.1. Notions

We will introduce the basic notions in our paper. Note that we use the lowercase letters (e.g., x) to represent scalars, the boldface lowercase letters (e.g., $\mathbf{x}$) to denote column vectors, and the boldface uppercase letters (e.g., $\mathbf{X}$) to denote matrices. A network is represented by an undirected unweighted graph $G = \{V, E\}$ where $V = \{v_1, \dots, v_n\}$ is the set of n nodes and $E \subseteq V \times V$ is the set of edges between nodes. The set of node v 's neighbors is defined as $\mathcal{N}(v) = \{u \in V | (u, v) \in E\}$, and $d(u)$ denotes the degree of node u . The one-hop egonet of node v_i is defined as $\mathcal{G}_i = \{V(g_i), E(g_i)\}$, where $V(g_i) = \{v_i\} \cup \{v_j \in V | (v_i, v_j) \in E\}$, and $E(g_i) = \{(u, v) \in E | u, v \in V(g_i)\}$. The extracted features of nodes are denoted as $\mathbf{X} \in \mathbb{R}^{n \times m}$, where m is the dimension of features. The aim of a network embedding method is to generate low-dimensional representation vectors (embeddings) for all the nodes, e.g., to learn a function $\mathcal{F} : V \rightarrow \mathbb{R}^k$, where $k \ll |V|$ is the dimension of embeddings. For clear clarification, we report the symbols in Table 1.

3.2. Model

Here we show the details of our proposed model *RESD*. As shown in Fig. 2, *RESD* consists of three parts: (1) the extraction process of role features, (2) a variational auto-encoder for composing the features into node representations, and (3) a regularizer based on node degrees for enhancing the representation ability of the learned embeddings.

3.2.1. Feature extraction

It is known that two nodes u and v tend to share the same role if they have similar structural identities. Though the adjacency matrix of graph contains all the topological information, it is hard to compare the structural identities of two nodes, as the adjacency matrix is generally sparse while the nodes may be far from each other or even disconnected. Thus, structural similarities cannot be learned directly from the adjacency matrix. Hence we need to extract compact higher-order structural features from adjacency relationships. Early works show many higher-order

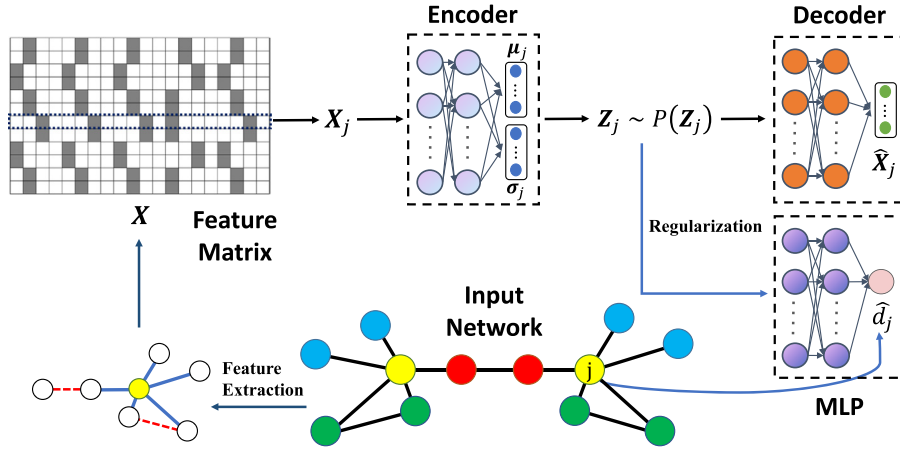


Fig. 2. An overview of proposed deep learning framework *RESD* (*RESD*): (1) extract features for all nodes; (2) generate embeddings $\mathbf{Z}$ via Variational Auto-Encoder; (3) enhance the embedding by regularizing with node degree.

feature extraction methods are effective on role discovery, such as motif-count-based features [12], degrees of multi-hop neighbors [31] and ReFeX [20]. All of these methods could be utilized as the feature extraction part of our framework *RESD*. Here, we adopt ReFeX [20] for its high efficiency and effectiveness, which generates higher-order structural features by aggregating simple local and egonet features in a recursive fashion. This methods extracts one local feature and five egonet-based features for each node. For a node v_i , the local feature is the node degree $d(v_i)$, and the egonet features based on $\mathcal{G}_i$ are defined as follows:

- Number of edges in $\mathcal{G}_i$: $f_1 = |E(\mathcal{G}_i)|$.
- Sum of degree for nodes in $V(\mathcal{G}_i)$: $f_2 = \sum_{u \in V(\mathcal{G}_i)} d(u)$.
- The proportion of edges in $\mathcal{G}_i$ to all edges within and leaving $\mathcal{G}_i$: $f_3 = f_1/f_2$.
- The proportion of edges leaving $\mathcal{G}_i$ to all edges within and leaving $\mathcal{G}_i$: $f_4 = (f_2 - f_1)/f_2$.
- The clustering coefficient for node v_i .

3.2.2. Variational auto-encoder

Because a lot of noise exists in real-world networks, the extracted higher-order structural features are not satisfactory enough for role discovery as they are always sensitive to the noise. Therefore, we utilize the Variational Auto-Encoder (VAE) [24] to encode structural features into node representations. VAE can reduce noise and capture nonlinear relationship between input features. Compared to ordinary Auto-Encoder (AE), it is effective for VAE to avoid over-fitting problem and improve the robustness of output embeddings. We use the Multi-Layer Perceptron model instead of Graph Neural Networks (GNNs) as our encoder, because the fact that GNNs always smooth the embeddings would negate extracted features.

In the VAE part of our proposed model *RESD*, given the extracted feature vector $\mathbf{X}_j$ of node v_j , the encoder is defined as follows:

$$\mathbf{h}_j^l = \tanh(\mathbf{W}^l \mathbf{h}_j^{l-1} + \mathbf{b}^l), l = 1, 2, \dots, L, \quad (1)$$

where $\mathbf{W}^l$ and $\mathbf{b}^l$ are the weight and bias of the l th layer, L is the number of hidden layers in encoder, and $\tanh(\cdot)$ is the hyperbolic tangent activation function. $\mathbf{h}_j^0 = \mathbf{X}_j$ is the input of encoder.

We assume the representation $\mathbf{Z}_j$ of node v_j follows a Gaussian distribution $\mathbb{P}_j(\mathbf{Z}_j)$, where all the elements of $\mathbf{Z}_j$ are independent and $\mathbb{P}_j(\mathbf{Z}_j) = \mathcal{N}(\mu_j, \text{diag}(\sigma_j^2))$. The parameters μ_j and σ_j of the distribution $\mathbb{P}_j(\mathbf{Z}_j)$ can be learned as follows:

$$\begin{aligned} \mu_j &= \mathbf{W}^\mu \mathbf{h}_j^L + \mathbf{b}^\mu, \\ \log \sigma_j &= \mathbf{W}^\sigma \mathbf{h}_j^L + \mathbf{b}^\sigma \end{aligned} \quad (2)$$

where $\mathbf{W}^\mu$, $\mathbf{b}^\mu$, $\mathbf{W}^\sigma$ and $\mathbf{b}^\sigma$ are respectively the weight and bias of the last layer of the encoder, with the previous parameters shared. We take the logarithm on σ to guarantee that the standard variance is always positive. Then we can obtain the node representation $\mathbf{Z}_j$ through a reparameterization trick [24], which allows us to compute the gradient for μ_j and σ_j :

$$\begin{aligned} \mathbf{Z}_j &= \mu_j + \sigma_j \odot \epsilon, \\ \epsilon &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{aligned} \quad (3)$$

where $\odot$ is the element-wise product, and ϵ follows a standard Gaussian distribution.

For the decoder part of *RESD*, we equally use a Multi-Layer perceptron model:

$$\hat{\mathbf{h}}_j^l = \tanh(\hat{\mathbf{W}}^l \hat{\mathbf{h}}_j^{l-1} + \hat{\mathbf{b}}^l), l = 1, 2, \dots, L, \quad (4)$$

where $\hat{\mathbf{W}}^l$ and $\hat{\mathbf{b}}^l$ are the weight and bias of the l th hidden layer in decoder. $\hat{\mathbf{h}}_j^0 = \mathbf{Z}_j$ is the input of the decoder and $\hat{\mathbf{X}}$ is the reconstructed feature matrix where $\hat{\mathbf{X}}_j = \hat{\mathbf{h}}_j^L$. Finally, we define the loss of VAE as:

$$\mathcal{L}_{VAE} = \|\mathbf{X} - \hat{\mathbf{X}}\|_2^2 = \sum_{j=1}^n \|\mathbf{X}_j - \hat{\mathbf{X}}_j\|_2^2 \quad (5)$$

3.2.3. Regularization

Algorithm 1: Algorithm for Eq. (7).

Input: A network $G = (V, E)$ and the weight of the degree-based regularizer

Output: Network representation $\mathbf{Z}$ for all the nodes

- 1: Extract higher-order structural features $\mathbf{X}$;
- 2: **Repeat**
- 3: **for** $j = 1$ to $|V|$ **do**:
- 4: Apply Eq. (3) to obtain $\mathbf{Z}_j$;
- 5: Apply Eq. (5) to compute the reconstruction error in VAE;
- 6: Apply Eq. (7) to compute the total loss $\mathcal{L}$;
- 7: Use back-propagate on $\mathcal{L}$ to update parameters;
- 8: **end for**
- 9: **Until** converge
- 10: Apply trained parameters to obtain the matrix $\mathbf{Z}$ of node representations;
- 11: **Return** $\mathbf{Z}$;

While the above VAE model has the ability to generate expressive representations, it runs the risk of losing some key structural

information when it estimates the distribution of input features. Therefore, it is crucial to add a constraint to preserve the key structural information into the embeddings.

From the criteria of traditional graph-based role discovery method such as structural equivalence and regular equivalence, it can be inferred that nodes playing the same role have similar degrees. Thus, degree is the most significant structural characteristic of a node, it is experimentally validated by many methods such as Demo-Net [45] and DRNE [23].

Similar to DRNE [23], we define the degree-based regularizer for our *RESD* as follows:

$$\mathcal{L}_{reg} = \sum_{j=1}^n (\log(d(v_j) + 1) - \text{MLP}(\mathbf{Z}_j))^2, \quad (6)$$

where $\text{MLP}(\cdot)$ is a Multi-Layer perceptron model with rectified linear unit activation $\text{ReLU}(\cdot)$.

We train our deep learning framework *RESD* by jointly minimizing the loss of feature reconstruction and degree-regularized constraint as follows:

$$\mathcal{L} = \mathcal{L}_{VAE} + \beta \mathcal{L}_{reg}, \quad (7)$$

where β is the weight of the degree-based regularizer. The whole algorithm is proposed in Algorithm 1.

3.3. Computational complexity analysis

The computational complexity of the proposed framework *RESD* partly depends on the higher-order structural feature extraction method. While a highly efficient feature extraction method is utilized, our framework's runtime is mainly attributable to the VAE part.

Here we take the example of ReFeX [20] used in this paper. Given a network G , let n denote the number of nodes, e denote the number of edges, r indicate the feature aggregation number of ReFeX, m be the dimension of extracted feature matrix $\mathbf{X}$, and k represent the dimension of node embeddings $\mathbf{Z}$. The total process of ReFeX takes $O(rm(e + nm))$ as each feature aggregation has higher computation than the previous aggregation. The representation for each node is obtained by VAE, of which computation complexity can be expressed as $O(m^2k)$, while the regularization takes $O(k^2)$ time. So the computation of embedding generation on the entire network is $O(nm^2k)$ for $k < m$. Since r is always very small and $r < k < m \ll n$, the whole computation of *RESD* is $O(rm + nm^2k)$. Therefore, our framework *RESD* has great potential for large-scale networks where it could take $O(nm^2k)$ time.

4. Experiments

4.1. Baselines

We compare our framework with both classic and role-based network embedding methods. We choose six state-of-the-art methods including node2vec [8], role2vec [12], struc2vec [31], RoIX [21], DRNE [23], and struc2gauss [48]. The dimension of embedding is set to 128 for all the baselines unless otherwise stated, while other parameters are set to default in their papers.

4.2. Datasets

We conduct experiments on several real-world networks with unweighted undirected edges. The datasets we use are listed as follows and the details are shown in Table 2:

Table 2

A brief introduction to datasets. The first six networks are used for role classification. Others are used for bot detection, where gl and ht denotes two languages: Galician and Haitian.

Dataset	# Nodes	# Edges	# Classes	Destiny (%)
Hospital	75	1,139	4	41.0450
Brazil	131	1,003	4	11.7792
Europe	399	5,993	4	7.5478
USA	1,190	13,599	4	1.9222
Actor	7,758	26,646	4	0.0886
Film	27,312	122,514	4	0.0328
Coauthor	146,092	400,863	3	0.0038
Dataset	# Nodes	# Edges	# Bots	Destiny (%)
gl-wiki-talk	7,935	19,887	12	0.0632
ht-wiki-talk	446	758	24	0.7638

- **Air-traffic networks** [31]: There are three networks, consisting of American, Brazilian, and Europe air-traffic networks (USA, Brazil, and Europe for short). In these networks, nodes represent airports and edges represent the existed flights between airports. The airports are labeled based on the volume of airplanes or people.
- **English-language film network** [49]: It is a film-director-actor-writer-network (Film for short). In Film network, node labels denote four types of identities: film, director, actor, and writer. And edges denote whether two nodes appear in the same Wiki page.
- **Actor co-occurrence network** [34]: It is an actor-occurrence network (Actor for short) extracted from the above Film network. In Actor network, nodes represent actors and are labeled based on their influences which are measured via the number of words in their Wiki pages.
- **Hospital ward dynamic contact network** [50]: This dataset contains the temporal network of contacts among patients, and health-care workers in a hospital ward in Lyon, France, from December 6, 2010 to December 10, 2010 (Hospital for short). The nodes are labeled by their identities: patient, medical doctor, administrative staff, and paramedical staff.
- **Aminer coauthor network**¹: This network represents the relationships of cooperation between authors. The original dataset contains 1,560,640 authors and 4,258,946 relationships, and we extract a connected subgraph where the word "system" is in their keyterms (Coauthor for short). Nodes represent authors and edges denote relationships, and the nodes are labeled by their h-indexes.
- **Wikipedia talk network** [51]: The network contains all the users and discussion from the inception of Wikipedia till January 2008. This network is separated into several subgraphs by the languages. Nodes in the network represent Wikipedia users and edge from node i to node j represents that user i edited a talk page of user j at least once. Most of the users are humans, while few of them are bots.

4.3. Experiment settings

For the part of feature extraction, we use ReFeX and set the number of feature aggregations to 3, and the number of bins is set to 5. For both encoder and decoder in VAE, we set the number of hidden layers to 2. And for degree-based regularizer, we use a 2-layer MLP model with ReLU activation function. We apply Adam SGD optimizer [52] with the learning rate of 0.001 and set the L_2 regularization with weight of 0.001 to avoid over-fitting. The batch size is set to 32 for most 50 epochs. In our later experiments, if not stated specifically, the dimension of embedding is set to 128, and β is set to 0.6.

¹ Available at <https://www.aminer.cn/panther>.

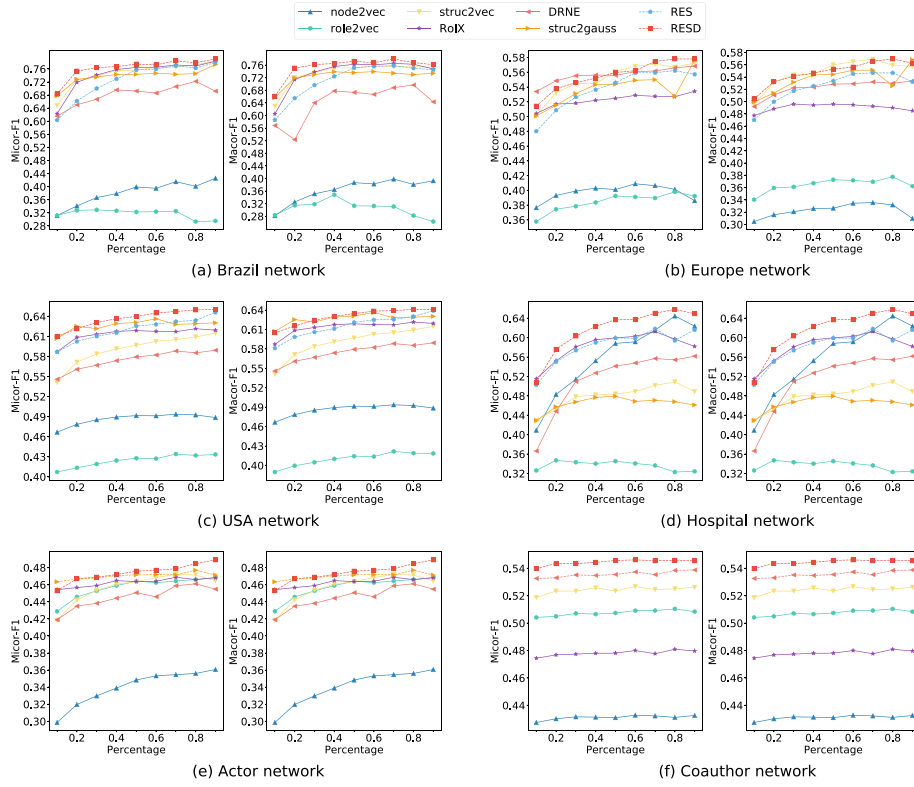


Fig. 3. The performance of different embedding methods on structural role classification. The F1-Micro and F1-Macro are computed on the average of running 20 times.

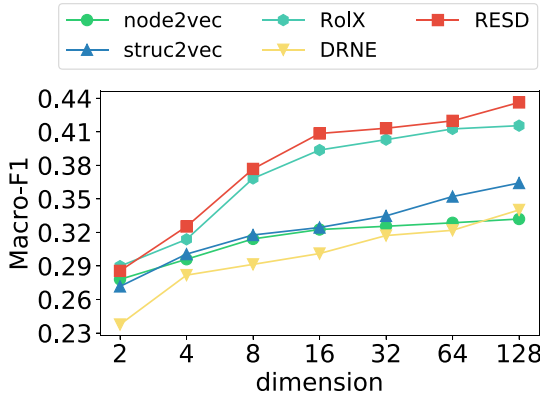


Fig. 4. The average Macro-F1 score of node classification on Film with different dimensions.

4.4. Role classification

We conduct the task of role-based node classification on seven real-world networks as shown in the top part of Table 2. In this experiments, we randomly different percentages of data to train a linear logistic regression classifier with embeddings inputted, while the rest are used for test. We run all the baseline methods and *RESD* for 20 times and compute their average F1 score on each network. To validate the effectiveness of the degree-preserving constraint, we also run a variant of *RESD* without the degree-based regularizer which is denoted as *RES*. The results are shown in Fig. 3.

It can be observed that the proposed framework *RESD* achieves best performance among all the networks, and the degree-preserving regularization can obviously promote the performance of representations. For the large-scale network (Coauthor), we find

Table 3

Precision at K selected users on bots detection task. OT denotes that method cannot obtain results within 48 h.

Method	gl-wiki-talk		ht-wiki-talk	
	50	100	50	100
node2vec	0.0083	0.0050	0.0625	0.0396
role2vec	OT	OT	0.0408	0.0350
struc2vec	0.0100	0.0067	0.1158	0.0808
RoIX	0.0200	0.0117	0.1142	0.0829
DRNE	0.0083	0.0067	0.0533	0.0500
struc2gauss	0.0150	0.0117	0.0917	0.0779
<i>RESD</i>	0.0200	0.0167	0.1225	0.0958

the scores become stable with the percentage of training samples increasing. The reason is that the number of nodes is large enough, and even small percentage of data can train the linear logistic regression classifier to get convergence. The results demonstrate the effectiveness of our methods. In addition, we choose a network with proper scale (Film) to evaluate the sensitivity to dimension of our model. The results are reported in Fig. 4. Note that the results of *struc2gauss* and *role2vec* on Film are not shown because they are too time-consuming and cannot generate embeddings within 48 hours. We observe that *RESD* obtains good enough result at the dimension of 16, which is applicable for other real-world datasets and downstream tasks.

4.5. Bot detection

In this section, we aim to explore whether our model can detect rare roles in some networks. We choose two Wikipedia talk networks where bots make up a small percentage of the total users. Thus it is much complex to detect bots on these networks. We first generate the representation of each user. For each bot, we compute the euclidean distance between it and

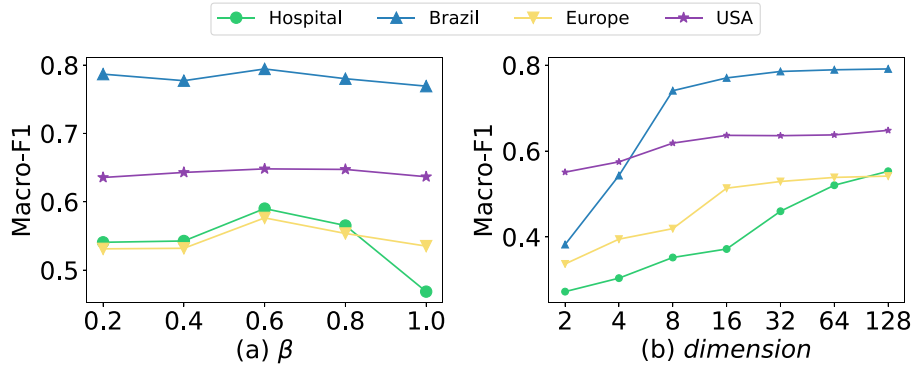


Fig. 5. Parameter sensitivity on several networks with respect to (a) the weight β of regularization, and (b) the dimension of embedding.

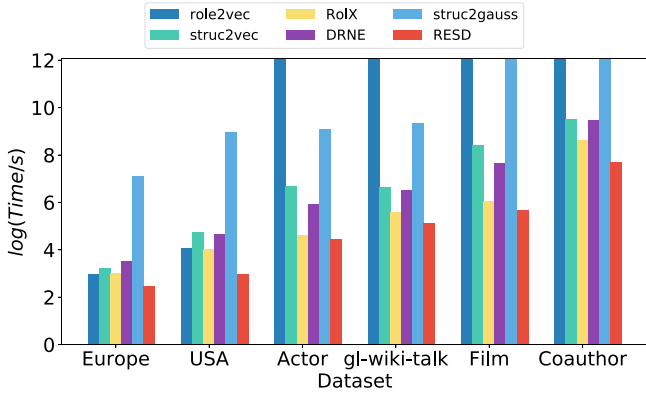


Fig. 6. Time of generating embedding on different networks. The full value means the method cannot obtain results within 48 h.

other users. Finally, we choose the K closest users to each bot and compute the average precision of bots. The results are shown in Table 3. Our framework *RESD* obtains the best performance. *RoIX* gets the closest results with our *RESD*, because we both leverage the ReFeX features. The difference is we apply VAE to capture nonlinear relationships and reduce the noise contained in the features, which can be achieved by Non-negative Matrix Factorization that *RoIX* uses.

This demonstrates that our method can extract more realistic role similarities, which can be applied for complex tasks like rare role detection.

4.6. Parameter sensitivity and computational complexity

The encoding process of VAE may lead to loss of key structural information such as degree, so we aim to seek which combination of parameters can capture structural identities of the original networks. To solve this problem, we conduct experiments on several networks with different parameter settings including dimension.

- Effect of β . Parameter β is the weight of degree-preserving regularizer. As shown in Fig. 5(a), with the increase of β , the Macro-F1 score of role classification increases. However, when $\beta > 0.6$, the score declines. Finally, we get the best result with $\alpha = 0.6$. It confirms that the regularization of node degree can supplement missing information in the process of encoding, but if the constraint is too strict, the representations will lose basic structural features.
- Effect of dimension. Fig. 5(b) shows how the dimension of embedding affects the performance. We observe that the performance improves with the increase of dimension, since

higher dimension can carry more structural information. When dimension exceeds 16, the results become stable. Because the network size of Hospital is too small and nodes are not labeled uniformly, it shows different trends. The results prove that our algorithm scales well with embedding dimensions.

Furthermore, we compare the runtime of embedding generation with baselines on these networks. As discussed in Section 3.3, the low computational complexity is one of the major issue in our method. We report the runtime of the baseline methods and our *RESD* in Fig. 6, in which the full value means that it cannot get results within 48 hours. The dimension of output embeddings for all these methods is set to 128. It can be observed that our *RESD* get the lowest runtime on all the networks. Though *RoIX* and *DRNE* have similar runtime to us, their performance on role classification is poorer. These results demonstrate the efficiency of our model.

4.7. Visualization

In this section, we compare *RESD* and *RES* with baselines on the task of network visualization. The Brazil network is selected due to its proper size and uniform labels of node roles. We apply t-SNE to reduce the dimension of embeddings to 2 for visualization. As shown in Fig. 7, each node is represented as a point, and the color denotes its role. Ideally, points in the same color should be close and those in different colors should be far away from each other. It is observed that *node2vec* and *role2vec* cannot extract role information well as the points in different colors are mixed up. The other methods achieve that points in the same color are clustered in varying degrees. However, *DRNE* may be over-fitting to one specific structure characteristic as the points are almost lined up. In addition, points of *RES* in the same color get closer than those of *RoIX*. And it is obvious that our *RESD* with degree-preserving regularization gets the best visualization result. The visualization experiment proves that our model effectively captures roles, and node degree is surely helpful for structural identities.

5. Conclusion

In this paper, we propose a novel framework *RESD* to generate high-quality role-based network embeddings. In detail, *RESD* has three parts: (1) the structural feature extraction part for capturing higher-order structural information; (2) the embedding generation part based on a Variational Auto-Encoder model for mapping structural similarities into embedding space and reduce the noise contained in extracted features; (3) the regularization part based on a degree-preserving constraint for preventing embeddings

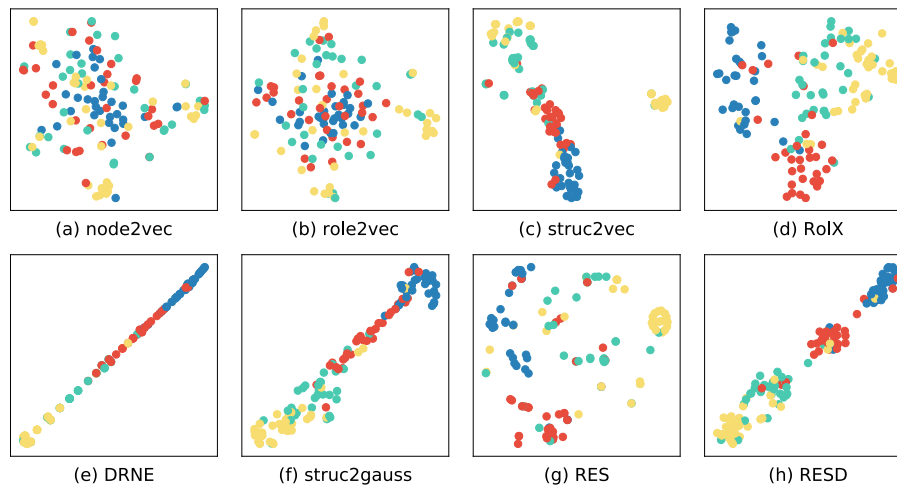


Fig. 7. Visualization of node representations on Brazil network in two-dimensional space. The label is mapped into color of point.

generated by VAE from losing key structural information. We take an example of *RESD* using ReFeX as the structural feature extraction part for framework introduction, theoretical analysis and experiments. The extensive experiments confirm the stability and effectiveness of *RESD*, and also prove that our framework scales well with network size and dimension.

There are also a few limitations. For example, the structural features in our model are too simple and the encoder ignores the network structure. And this model can only be applied on unweighted undirected graph. For future works, we plan to leverage graph neural network (GNN) to encode features while extracting complex relationships between nodes. We also hope to design a novel deep-learning framework to generate more expressive structural features than hand-made works.

To our best knowledge, most of role-based network embedding methods focus on static networks, and they ignore the additional attributes on nodes or edges. In order to understand the influences of roles better, we will further explore dynamic networks and introduce attributes to learn more effective representations, which might be attractive future works.

CRediT authorship contribution statement

Wang Zhang: Data curation, Writing - original draft, Writing - review & editing, Software. **Xuan Guo:** Visualization, Writing - original draft. **Wenjun Wang:** Conceptualization, Supervision. **Qiang Tian:** Validation, Writing - review & editing. **Lin Pan:** Writing - review & editing. **Pengfei Jiao:** Writing - original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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